PRODUCT REQUIREMENTS DOCUMENT (PRD)

Last-Mile Dynamic Technician Dispatcher

Intelligent On-Demand Service Marketplace & Dispatch Optimization Platform

Document Information	**Details**
Version	1.0
Status	Initial Draft – For Review
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Date	02 November 2026
Project Type	Full-Stack Web Application
Business Model	Multi-Sided Service Marketplace
Technology Stack	Next.js, Node.js, Express.js, MongoDB

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1. EXECUTIVE SUMMARY

The Last-Mile Dynamic Technician Dispatcher is an intelligent, location-based service marketplace designed to connect customers with independent technicians and service professionals for on-demand services, including Plumbing, HVAC and Urgent Delivery.

The platform aims to simplify service discovery, technician matching, job assignment and real-time service coordination through an intelligent dispatch optimization engine.

Unlike conventional booking platforms that primarily rely on geographical proximity or manual assignment, the proposed system evaluates multiple factors before recommending a technician, including:

- Real-time location and estimated travel time.
- Technician skills, qualifications and availability.
- Spare-parts and equipment inventory.
- Customer priority and Service Level Agreement (SLA) deadlines.
- Technician workload and service reliability.

The system will identify eligible technicians, rank them based on configurable optimization criteria and send job offers to suitable candidates. Technicians can accept or reject offers, while rejected or expired offers trigger automatic reassignment.

The platform will provide three primary interfaces: Customer Application, Technician Dashboard and Admin/Dispatcher Dashboard.

The proposed revenue model includes a platform fee of **$1.50 per successfully completed dispatch ticket**, subject to final business confirmation.

2. PROBLEM STATEMENT & PRODUCT OBJECTIVES

2.1 Problem Statement

Traditional service booking and dispatch systems face several operational challenges:

- **Inefficient Assignment:** Selecting technicians based only on proximity may result in unsuitable assignments.
- **Inventory Unavailability:** Technicians may lack the spare parts or equipment required to complete a service.
- **Travel-Time Uncertainty:** Traffic and road conditions affect technician arrival times.
- **SLA Violations:** Poor prioritization and delayed assignments can result in missed service deadlines.
- **Manual Reassignment:** Technician rejection, cancellation or unavailability creates additional operational overhead.
- **Limited Visibility:** Customers and administrators may lack accurate information about job progress and technician location.

2.2 Proposed Solution

The platform will introduce an intelligent dispatch system that evaluates technician eligibility, inventory readiness, location, travel time and SLA requirements before generating assignments.

2.3 Product Objectives

1. Automate technician matching and job assignment.
2. Reduce technician travel time and unnecessary service delays.
3. Ensure required spare parts and equipment are available before assignment.
4. Improve SLA compliance through priority-based dispatch.
5. Enable automated reassignment when technicians reject or cannot fulfill requests.
6. Provide real-time service tracking and operational visibility.
7. Establish a scalable marketplace for multiple on-demand service categories.

3. TARGET USERS & PRODUCT SCOPE

3.1 Target Users

1. Customers

Individuals or businesses requiring on-demand services.

Key capabilities:

- Register and manage profiles.
- Browse Plumbing, HVAC and Urgent Delivery services.
- Create service requests and provide location details.

- View pricing estimates and quotations.
- Track assigned technicians.
- Make payments and submit ratings.

2. Independent Technicians / Service Professionals

Plumbers, HVAC technicians and delivery partners registered on the platform.

Key capabilities:

- Register and complete professional verification.
- Manage skills, service areas and availability.
- Maintain spare-parts inventory or delivery equipment details.
- Receive, accept or reject job offers.
- Update job status and location.
- Submit completion evidence and view earnings.

3. Admin / Dispatcher

Platform operators responsible for managing marketplace operations.

Key capabilities:

- Manage customers and technicians.
- Verify technician profiles and documents.
- Monitor bookings and active dispatches.
- View technician locations and SLA performance.
- Manually override assignments when required.
- Manage payments, disputes and operational reports.

3.2 Product Scope

In Scope – Initial MVP

- Role-based authentication and authorization.
- Customer and technician onboarding.
- Technician verification and availability management.
- Service catalogue and booking management.
- Location-based technician discovery.
- Technician skills and inventory management.
- Intelligent dispatch and candidate ranking.
- Job acceptance, rejection and reassignment.
- Map-based tracking and ETA estimation.

- SLA monitoring and notifications.
- Job completion and verification.
- Payment processing and platform fee calculation.
- Ratings, reviews and admin dashboard.

Out of Scope – Initial MVP

Native mobile applications, advanced machine-learning prediction, multi-city fleet optimization, automated warehouse management and complex loyalty programs.

4. CORE PRODUCT WORKFLOWS

4.1 Customer Booking Workflow

1. Customer registers or logs in.
2. Selects a service category and describes the requirement.
3. Provides service location and relevant details.
4. Reviews estimated pricing or quotation requirements.
5. Confirms the service request.
6. System initiates intelligent technician matching.
7. Customer receives assignment and ETA updates.
8. Customer tracks the technician.
9. Service is completed and verified.
10. Payment is finalized and customer can submit a review.

4.2 Technician Workflow

1. Technician registers and submits professional details.
2. Administrator verifies and approves the profile.
3. Technician configures skills, service areas and availability.
4. Technician maintains spare-parts inventory or delivery capacity.
5. Receives a suitable job offer.
6. Accepts or rejects the offer.
7. Travels to the customer and updates job status.
8. Completes the service and submits required evidence.
9. Inventory consumption and job records are updated.
10. Technician can view earnings and completed jobs.

4.3 Intelligent Dispatch Workflow

Customer Request → Requirement Analysis → Technician Eligibility Filtering → Candidate Ranking → Job Offer → Technician Acceptance → Assignment Confirmation → Live Tracking → Job Completion.

If a technician rejects an offer, does not respond within the configured time or becomes unavailable, the system evaluates the next eligible candidate and initiates reassignment.

5. INTELLIGENT DISPATCH & OPTIMIZATION

The dispatch engine is the core component of the platform. It aims to select the most suitable technician rather than simply the nearest technician.

5.1 Eligibility Filtering

Before ranking candidates, the system will verify mandatory conditions:

- Technician account is approved and active.
- Required service skills and qualifications are available.
- Technician is currently available.
- Technician is within the configured service area.
- Required spare parts are available for Plumbing and HVAC jobs.
- Delivery partners meet vehicle and package-capacity requirements.
- No conflicting confirmed assignment exists.

Technicians failing mandatory conditions will be excluded.

5.2 Candidate Ranking

Eligible technicians will be ranked using a configurable weighted scoring model.

Optimization Factor	**Initial Weight**
Location and travel-time suitability	25%
SLA feasibility	30%
Skills and experience	15%
Workload and availability	10%
Inventory / equipment readiness	10%
Reliability and ratings	10%
Total	**100%**

The scoring model will use normalized values between 0 and 100. These weights are initial design assumptions and will be refined through testing.

5.3 Dynamic Reassignment

The system will support automatic reassignment when:

- A technician rejects or ignores an offer.
- A technician cancels an accepted job.
- A technician becomes unavailable.
- A significant delay creates an SLA risk.

Reassignment after travel has started may require dispatcher approval to avoid unnecessary disruption.

5.4 Service-Specific Requirements

Service	Special Dispatch Requirements
Plumbing	Plumbing skills, tools, pipe fittings and spare parts
HVAC	HVAC specialization, qualifications and AC components
Urgent Delivery	Vehicle type, package capacity, pickup/drop-off location and delivery deadline

6. BUSINESS MODEL & BUSINESS RULES

6.1 Proposed Business Model

The platform will operate as a multi-sided marketplace connecting customers with independent service professionals.

The platform will facilitate service discovery, booking, technician matching, dispatch coordination, payment processing and service verification.

Proposed Pricing Strategy:

- Standard services: Platform-defined base prices or estimated price ranges.
- Complex services: Technician-generated quotations requiring customer approval.

Proposed Revenue Model:

The project brief specifies a fee of $1.50 per successfully completed dispatch ticket.

Platform Revenue = Completed Eligible Tickets × $1.50

The exact fee payer, currency and settlement structure require business confirmation.

6.2 Key Business Rules

- Only verified technicians can receive job offers.
- Mandatory skills and inventory requirements cannot be bypassed by a high dispatch score.
- Technicians control their availability and can accept or reject offers.
- Inventory is reserved after assignment confirmation and adjusted after actual consumption.
- Delivery capacity requirements replace spare-parts requirements for delivery jobs.
- SLA deadlines must be recorded and monitored.
- Completed tickets must generate applicable platform fees only once.

- Customer and technician information must be accessible only to authorized users.

7. TECHNOLOGY STACK & NON-FUNCTIONAL REQUIREMENTS

The project will follow the organization's standardized technology stack.

Layer	Technology	Purpose
Frontend	Next.js	Responsive customer, technician and admin interfaces
Backend	Node.js + Express.js	APIs and business logic
Database	MongoDB	Users, bookings, inventory and transactions
Real-time Communication	Socket.IO	Live tracking and status updates
Maps	Leaflet + OpenStreetMap	Map visualization and location display
Routing	OSRM or equivalent	Route and travel-time estimation
Authentication	JWT / Secure Sessions	Authentication and authorization
Payments	Approved Payment Gateway	Customer payments and settlements
Version Control	Git + GitHub	Collaboration and source management

Traffic-aware ETA will require an appropriate traffic-data provider; standard routing alone does not guarantee live traffic information.

Non-Functional Requirements

- **Performance:** Common API operations should target response times below 500 ms under normal test conditions, excluding external service latency.
- **Scalability:** The architecture should support increasing users, technicians and service requests.
- **Security:** Implement secure authentication, role-based authorization, input validation, rate limiting and protected file uploads.
- **Reliability:** Prevent duplicate assignments, payment processing and inventory deductions.
- **Usability:** Provide responsive, intuitive and consistent interfaces.
- **Maintainability:** Follow modular architecture, reusable components and clean coding practices.
- **Privacy:** Protect personal information, payment records and technician location data.

8. SUCCESS METRICS & ACCEPTANCE CRITERIA

The platform's success will be evaluated using the following KPIs:

KPI	Measurement
Dispatch Success Rate	Percentage of requests successfully assigned
Average Assignment Time	Time required to confirm a technician
Average Arrival Time	Time between assignment and technician arrival
SLA Compliance Rate	Percentage of jobs completed within SLA
Technician Acceptance Rate	Percentage of offers accepted
Reassignment Rate	Percentage of jobs requiring reassignment
Job Completion Rate	Percentage of confirmed bookings completed
Customer Satisfaction	Average post-service rating
Platform Revenue	Revenue from eligible completed tickets

MVP Acceptance Criteria:

The initial release should demonstrate that:

1. Customers can create and track service requests.
2. Technicians can register, verify their profiles and manage availability.
3. Inventory and service requirements are validated.
4. The dispatch engine filters and ranks eligible technicians.
5. Technicians can accept or reject job offers.
6. Rejected or expired offers trigger reassignment.
7. Job status and relevant location updates are available.
8. SLA performance and completion records are maintained.
9. Payments and platform fees are recorded without duplication.
10. Administrators can monitor and manage marketplace operations.

9. ASSUMPTIONS, RISKS & FUTURE ENHANCEMENTS

9.1 Current Assumptions

- The product operates as an independent-technician marketplace.
- Initial services include Plumbing, HVAC and Urgent Delivery.
- Technician-managed inventory is used for repair services.
- The initial dispatch mechanism uses sequential job offers and technician acceptance.
- A $1.50 fee applies to eligible completed dispatch tickets.

- The initial launch will focus on a limited geographical area.
- SLA targets and offer-expiration times will be configurable.

9.2 Key Risks and Mitigation

Risk	Mitigation
Inaccurate technician location	Validate location updates and timestamps
Incorrect inventory data	Inventory reservations and verification
Technician rejection	Automated reassignment
Traffic uncertainty	Integrate suitable routing and traffic providers
Payment failures	Transaction reconciliation and retry mechanisms
SLA violations	Deadline monitoring and escalation
Unauthorized access	Role-based security and audit logging

9.3 Future Enhancements

- Advanced multi-job route optimization.
- Predictive SLA risk analysis.
- Demand forecasting and geographical heatmaps.
- Native Android and iOS applications.
- Advanced technician performance analytics.
- Automated inventory replenishment recommendations.
- Multi-city and multi-region expansion.
- AI-assisted operational insights.

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10. PENDING STAKEHOLDER DECISIONS

The following items will be finalized through stakeholder discussion and team brainstorming:

1. Exact fee payer and settlement model for the $1.50 dispatch fee.
2. Initial launch geography.
3. Service-specific response, arrival and completion SLA targets.
4. Final payment gateway and traffic-data provider.
5. Cancellation, refund and dispute policies.
6. Final dispatch scoring weights and offer-expiration duration.